ONE DAY STATE LEVEL SEMINAR ON

'EMERGING TRENDS IN MEDICINAL CHEMISTRY'

10 March 2015

A REPORT

INTRODUCTION

The Departments of Chemistry and Bio-chemistry of St. Philomena's college (Autonomous), Mysuru conducted a one-day state level seminar funded by University Grants Commission (under college with Potential for Excellence scheme) on 10th March 2015 on the topic 'Emerging trends in Medicinal Chemistry' in the college campus.

The theme of the seminar has been selected exclusively as medicinal chemistry is one of the thrust areas of research and is gaining importance day by day both in the industrial and medicinal fields all over the world.

OBJECTIVES

- To provide a platform for students and young researchers in the field of Medicinal chemistry and allied areas to present their work.
- To appreciate the role of synthetic Organic Chemistry in Medicine.
- To expose students and researchers by recent advancements in drug developments in the field.
- To facilitate exchange of ideas between experts and students.
- As a modest endeavour to bring together academicians, researchers and students from across the state under one roof to this urban part of the state to share their views, scientific thoughts and experiences with each other.

PRE-SEMINAR PREPARATIONS

The departments (both Chemistry and Bio-chemistry) in their meeting dated 14th February 2015 had decided to from the following committees for the smooth conduct of the seminar.

Following is the details of the various committees formed.

1. **PATRON:** Rev. Fr. Leslie Moras
2. **ORGANISING SECRETARY :** Dr. Ravi J.D.Saldanha , Associate Professor,
3. **CONVENER :** Prof. Agnes D' Souza , Associate Professor Department of Chemistry

4. ORGANISING COMMITTEE :

1. Rev. Fr. Leslie Moras, Principal
2. Rev.Fr.Bernard Prakash Barnis, Vice Principal
3. Prof. G. Rajalakshmi, Dean of Science and HOD of Physics
4. Dr. Alphonsus D' Souza, HOD , Department of chemistry and Bio-Chemistry
5. Prof. Agnes Sylvia D' Souza, Associate Professor , Department of Chemistry
6. Prof. Britto Dominic Rayan, Associate Professor, Department of Chemistry
7. Miss Lydia Vandana, Assistant Professor, Department of Bio-Chemistry
8. Miss Jennie Prathibha, Assistant Professor, Department of Bio-Chemistry
9. Prof. Cecelia N .A .D'Souza, HOD , Department of Zoology
10. Prof. Lizzy Cyriac, HOD , Department of Botany
11. Prof. Vidyadhar Sanjay Nair, HOD , Department of Mathematics

5. REGISTRATION COMMITTEE:

1. Prof. Agnes S. D' Souza, Associate Professor
2. Prof. Kunal , Assistant Professor, Department of Electronics

6. BOARDING & LODGING COMMITTEE:

Mr. Avinash Rajashekhara, Assistant Professor, Department of Criminology

7. INVITATION, PRESS & PUBLICITY COMMITTEE :

Dr. Alphonsus D'Souza, HOD, Department of Chemistry and Bio-Chemistry

As the pre-seminar initiative, the brochure was sent to various colleges of Karnataka state through post as well as e-mail. The seminar invitation had been sent to all the registered delegates. Nearly 250 delegates registered and actively participated in this seminar. As more focus was laid on the participation of our own college UG and the PG students of Chemistry & Bio-Chemistry, a large number of students attended this seminar. A registration fee of Rs 250/- for faculty and Rs 100/- for research scholars/students was collected. A seminar kit comprising of a folder, a pen and a scribbling pad was provided to all registered delegates and special invitees.

PROCEEDINGS

THE INAUGURAL SESSION

The programme of the inaugural session began at 10 AM with an introduction given by Miss Preetha D' Souza, faculty of PG Chemistry department who was also the Master of ceremonies for the entire day's proceedings. A few students of B.Sc of the college rendered the invocation followed by a semi-classical welcome dance. The chief guest of the inaugural session was Dr

Balakrishna Kalluraya, Professor of organic chemistry, DOS & Research in Chemistry, University of Mangalore who was also invited to deliver the keynote address. Rev. Fr. Leslie Moras, Principal of the college presided over and Rev Fr Bernard Prakash P Barnis, Vice-Principal of the college was the Guest of honour. Other dignitaries viz Prof Rajalakshmi, HOD of Physics Department & Dean of Sciences, Prof. Alphonsus D'Souza , HOD of Chemistry & Bio-Chemistry, Dr Ravi J D Saldanha, Organizing secretary of the seminar were also present on the Dias.



Prof Agnes S D'Souza, Associate Professor & convener of the seminar while briefing about the seminar highlighted the significance of emerging areas in the Medicinal Chemistry and also its importance to the research areas in modern times and also justified the relevance of the themes of the seminar. Dr Alphonsus D'Souza welcomed the guests and the gathering. The seminar was then officially inaugurated by lighting the lamp by dignitaries on the dias. In his inaugural address, Prof Balakrishna said, "Chemistry is the mother of science. Chemistry gives us life. It is a beautiful subject which has wide applications in our day to day lives". He gave examples of common salt, table sugar, water, pesticides, fertilizers and drugs to explain the importance of chemistry. Rev. Fr. Leslie Moras, Principal of the college, who presided over the function,

While speaking about the importance of conducting such seminars said, "Classroom teaching is one mode of learning. The other modes of learning are seminars, symposia, workshops, science exhibitions etc. One can understand the subject better by attending seminars".

Soon after the presidential remarks, Prof. Balakrishna Kalluraya was invited to deliver the keynote address. He spoke on 'Drug discovery and Drug Development' Prof. Kalluraya explained about primitive medicine, Indian medicine (Ayurveda and Siddha), Allopathy, Homeopathy and Unani through his power point presentation. While speaking about different sources of drugs, he said that drugs (medicines) can be prepared from chemical synthesis, micro-organisms, minerals, plants and animals. The concept of studying the drug and the effects of drugs like chloral hydrate, aspirin etc were discussed in his presentation. He also presented a

brief history of the drug discovery. At the end, there was a brief question-answer session. Prof. Kalluraya was honoured with a memento. Dr Ravi J D Saldanha proposed a vote of thanks.

TECHNICAL SESSION -1

The first technical session began at 11.03AM soon after the tea-break. Dr Hanumanthachar Joshi, Principal, Sarada Vilas Pharmacy College, Mysore was the resource person. Dr Joshi spoke on the topic 'Phyto Pharmacotherapy for neurodegenerative disorders-challenges and opportunities for medicinal chemists'. In his PowerPoint presentation, he narrated different types of neurological disorders such as Alzheimer's. He mentioned about the symptoms of the disease and possible precautionary measures to be taken for preventing diseases like Parkinson's, Alzheimer's etc



After giving basic ideas of the diseases, he discussed in detail about the scope of chemistry in curing these diseases.

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He later discussed about the drugs that are currently used in the treatment of these diseases and demonstrated their mode of working. He then displayed a few slides of conducting experiments on animals like mice having conditions of Alzheimer's and how using drugs they were able to improve their memories. At the end he said that one should not forget to extend mental and moral support to patients suffering from Alzheimer's and Parkinson's diseases. The session ended with a brief interactive sessions with the participants.

TECHNICAL SESSION -2

When the participants reassembled after a sumptuous lunch at 2.00PM Dr. Shivaramayya K, Senior Manager, BIOCON Ltd, Bangalore spoke on the topic 'Drug discovery'. In his presentation, he said that drug discovery is possible only with the knowledge of chemistry, Bio-chemistry and Bio-Technology. He further gave a few tips regarding the factors to be noted while preparing a drug such as

i. Choosing a lead molecule
ii. Identifying a bioassay
iii. Location and design
iv. Structural determination
v. Isolation
vi. Purification etc.

He stressed on the fact that one must be able to manufacture drugs in such a way that it only reacts at specific areas which should be cured without damaging other cells.



He was then able to convince the audience regarding the risk factors involved and the amount of waiting period required in the process of developing a new drug. He showed a few slides depicting the phases involved in the middle of each process. The chances and reasons for failure of a few new drugs was also discussed in detail.

He emphasized the need of a few young chemists in the field of medicinal Chemistry to use their innovative ideas. At the end of this session the speaker answered a few questions from the anxious participants.

TECHNICAL SESSION-3

The theme of this session was ‘Crystallinity and its impact on Medicinal Chemistry’. The resource person was Dr. Shridhara Bhat K, CEO, Shcub Laboratories, Bangalore. He spoke about the various processes involved in crystallization and re crystallization, purification, particle size and R&D experiments. He also gave a brief introduction about how the science of crystallinity can be applied in Medicinal chemistry especially for the synthesis of drugs. He then spoke on some of the basic properties of crystals and their structures, orientation etc.



He later explained how the activity of a drug is affected when it is converted to its crystalline form and how the science of crystallinity becomes applicable in drug synthesis. Later in his talk, he spoke about the properties of a drug in its crystalline form. He made it clear that use of natural product is considered as favorable and less toxic method for any commercial drug designing company. He also spoke about the small but important aspects of drug synthesis like time management, different and easy methods for purification etc. The session concluded by discussing those of the formalities and procedures to be adopted for obtaining the patent of drugs and how the health and moral aspects of business are taken lightly in the field drug synthesis.

VALEDICTORY

The valedictory session began at 4.03PM Dr H. R. Shivakumar, Special officer, Vishweswarayya Technical University Regional centre, Mysuru was the chief guest.

Rev Fr Leslie Moras, Principal presided over. Dr Ravi J D Saldanha, organizing secretary formally welcomed all the guests and participants. This was followed by a feed back session in which three participants gave their feedback about the seminar .Yasasri Prabash a B Sc VI semester Bio-Chemistry student gave a brief summary of the day's proceedings of the seminar. He presented a brief report and summed up the deliberations of the seminar. In his valedictory address Dr Shivakumar explored the various opportunities in the medicinal chemistry field and scope for research.



Rev Fr Leslie Moras, Principal of the college in his presidential address gave the concluding remarks of the day-long seminar proceedings stretching the importance of having more of this type of seminars in order to be resourceful in all fields. Miss Lydia Vandana, Assistant Professor of Bio-chemistry thanked each and everyone who was directly and indirectly involved in the smooth conduct of the seminar, Participants were informed to collect their certificates after the programme.

The one-day seminar ended with the National Anthem.

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